Abhishek Kumar Singh

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Examination	Degree	Institute	Year	CPI/%
Post Graduation (CSE)	MS By Research	IIT Bombay	2022-25	8.48
Graduation (CSE)	BTech	REC Sonbhadra	2018-22	8.13

WORK EXPERIENCE & COLLABORATION

- **Qualcomm, Bengaluru India** ***Open Source at-*** Git-Hub Link *Machine Learning Engineer*
  Team : ***Cloud ML (Model onboarding)*** | LLMs | Agents | Efficient Transformers *(July'25 - Present)*
  - Delivered **60% compile-time speedup** by adding subfunction support in Efficient Transformers.
  - Implemented **layer-wise execution** for large models (for example, Kimi K2.5), reducing peak memory to **<91 GB RAM** and enabling export of a 1 trillion-parameter model in **1.5 hours**; **patent filing in progress**.
  - Performed a detailed analysis of **torch.compile** versus **torch.onnx.export** for model onboarding and deployment reliability.
  - Built **DocQGenie**, an agentic framework that auto-generates repository documentation from GitHub codebases (internal hackathon project).
- **IBM Research, Bengaluru India** [***Published*** *at : NAACL 2025*] Paper Link *Research Intern AI*
  Mentor: ***Vishwajeet Kumar, Rudra Murthy, Ashish Mittal, Jaydeep Sen*** | LLMs | RAG *(May'24 - Aug'24)*
  - Submitted **2 papers** during the internship and released a **large multilingual QA benchmark** to evaluate context-grounded question-answering in open-source multilingual LLMs.
  - Conducted a comparative study of **multilingual-query** and **translation-based** pipelines for Indian language QA systems.
  - Initiated a multi-LLM composition framework that combines model-specific strengths to improve multilingual QA performance (later extended as part of M.S. thesis work).

RESEARCH PROJECTS & PUBLICATIONS

- **EIGEN: Expert-informed joint learning aggregation for high-fidelity information extraction from document images** [***Published*** *at : ML4H* (co-located with NeurIPS 2023)] Paper Link *M.S. Thesis*
  Guide: ***Prof. Ganesh Ramakrishnan*** | NER | Information Extraction | SPEAR | PyTorch *(Jan'23 - Jan'25)*
  - Developed **CAGE**, an unsupervised token classification approach that reduces dependency on labeled training data.
  - Fine-tuned **LayoutLM** as a strong baseline for document-level token classification tasks.
  - Built a **joint learning framework** using **SPEAR**, leveraging semi-supervised learning to achieve state-of-the-art results.
  - Proposed a **joint LLM fine-tuning strategy** with data programming, improving extraction accuracy while reducing labeling effort and cost.
  - Automated **labeling-function generation** and developed a deployable workflow suitable for practical usage.
- **Composition of LLMs for Multilingual Question Answering** [*Collaboration with IBM Research*] *M.S. Thesis*
  Guide: ***Prof. Ganesh Ramakrishnan*** | Multilingual NLP | LLM Composition | RAG *(May'24 - Jan'25)*
  - Developed a composition strategy to combine multiple LLMs for improved multilingual question-answering performance.
  - Implemented ensemble strategies that leverage complementary strengths of diverse language models across 10+ Indic languages.
  - Achieved performance gains by combining model outputs using language-specific routing and selection heuristics.
  - Built a comprehensive evaluation framework for multilingual QA systems, with emphasis on zero-shot generalization.

PROJECTS

- **Fair Diffusion: Mitigating Bias in Conditional Diffusion Models** *Advanced Machine Learning*
  Guide: ***Prof. Sunita Sarawagi*** | Generative Models | Debiasing | PyTorch *(Spring'23)*
  - Identified and mitigated bias issues in Conditional Denoising Diffusion Probabilistic Models.

- Applied **supervised contrastive loss** to reduce bias in generated images.
- Conducted extensive experiments on correlated and imbalanced MNIST datasets with quantified fairness metrics.

- **Fact-Checking Claims from Wikipedia** *Organization of Web Information*
  Guide: ***Prof. Soumen Chakrabarti*** | RAG | Retrieval *(Spring'24)*
  - Implemented an embedding-based retrieval system using **Sentence Transformer** for Wikipedia data, achieving a Recall@100 of **0.707**.
  - Enhanced retrieval precision by integrating **cross-encoder** re-ranking and **Named Entity Recognition** (NER), reducing false positives by **28%** across claim types.
- **Reverse Codex: Automatic Code Comment Generation** *Speech, NLP, and the Web*
  Guide: ***Prof. Pushpak Bhattacharyya*** | Bi-LSTM | Sequence-to-Sequence | PyTorch *(Autumn'22)*
  - Developed an encoder-decoder model with attention for automatic Java code comment generation.
  - Implemented **Structured-Based Traversal (SBT)** for precise Abstract Syntax Tree (AST) navigation and code representation.
  - Achieved BLEU scores competitive with state-of-the-art models, demonstrating model efficacy.
- **OCR to Speech Pipeline: End-to-End Document-to-Audio System** *Deep Learning for NLP*
  Guide: ***Prof. Pushpak Bhattacharyya*** | OCR | Summarization | Machine Translation | TTS *(Spring'23)*
  - Fine-tuned pre-trained OCR models on the DocBank dataset and integrated PEGASUS for abstractive summarization.
  - Built transformer-based neural machine translation on the CFLIT Hindi dataset for accurate cross-lingual translation.
  - Implemented text-to-speech synthesis to complete the end-to-end document-to-audio pipeline.
- **Breast Cancer Classifier** *Foundation of Machine Learning*
  Guide: ***Prof. Preethi Jyothi*** | CNN | SVM | Ensemble| PCA | Keras *(Autumn'22)*
  - Conducted model comparisons across **SVM**, ensemble methods (**XGBoost**), and **CNN**.
  - Employed **PCA** for dimensionality reduction on the **BreakHis** dataset.
  - Determined CNN as the **optimal** performer.
- **Simulation of Mining Attacks on P2P Cryptocurrency network** *Introduction to Blockchains, Cryptocurrencies*
  Guide: ***Prof. Vinay Ribeiro*** | Blockchain | Simulation *(Spring'24)*
  - Designed a **discrete-event** simulator for a **peer-to-peer** cryptocurrency network for 10-100 users.
  - Simulated **selfish mining** scenarios with **dual attackers** to assess their impact on blockchain integrity.
  - Evaluated the influence of varying miner **hash power** and **network delays** on transaction throughput and efficiency.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, MATLAB, bash
- **Development:** HTML, CSS, PHP, GitHub Actions
- **Tools & Libraries:** Git, Scikit-learn, PyTorch, Keras, Hugging Face Transformers, ONNX, LaTeX

ACHIEVEMENTS AND EXTRA-CURRICULAR

- Achieved a **99.47** percentile in GATE 2022.
- First author of a paper published in **ML4H 2023**, co-located with NeurIPS 2023, focusing on information extraction from document images.
- Achieved **Top Performer** in **TCS NQT** 2022.
- **Hobbies** : Cricket, Travelling, Chess, Music, Web Series